

Nicole Ye's Portfolio

[Linkedin](#) | [Nicole Ye's Resume](#) | 647-507-7822 | n7ye@uwaterloo.ca | [Github](#)

If you have any inquiries, please don't hesitate to reach out!

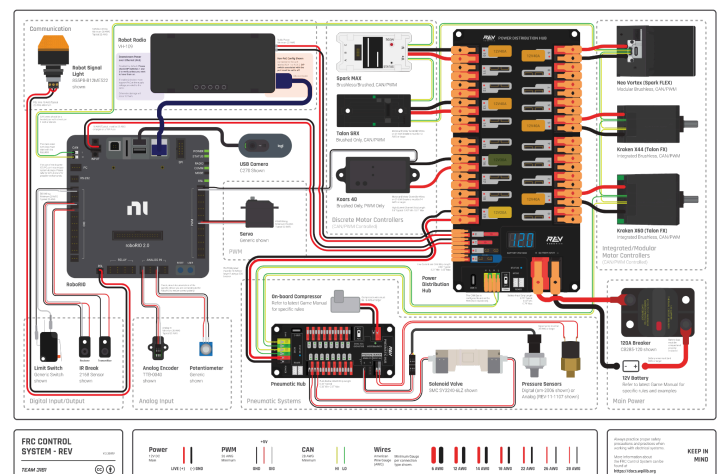
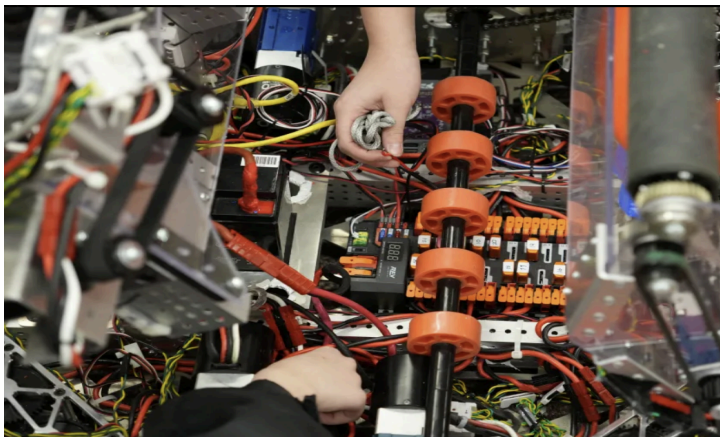
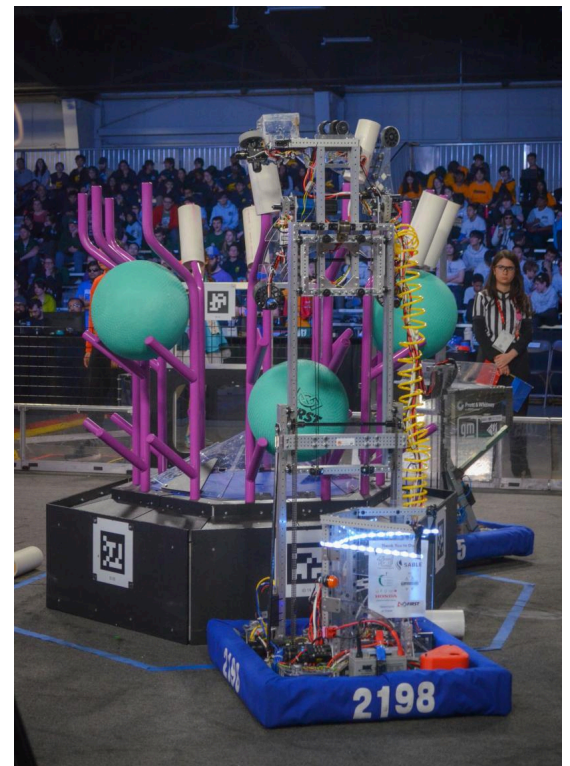
SKILLS

- **Languages:** Java, Python, Javascript, HTML/CSS, C++
- **Programs:** Onshape, Solidworks, Visual Studio Code, Microsoft Excel/Teams, Slack, Discord
- **Electrical Skills:** Soldering, Multimeter, Cable Management, Crimping, Testing
- **Soft Skills:** Project Management, Critical Thinking, Organization, Customer Service

Team 2198 Paradigm Shift Robotics Electrical System

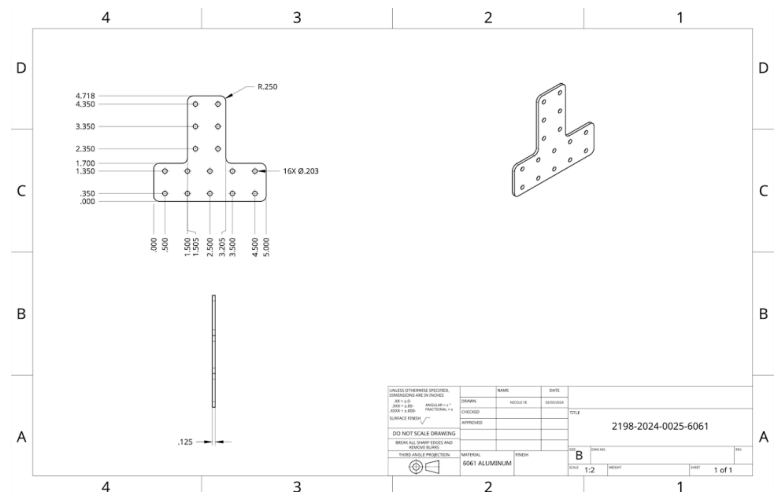
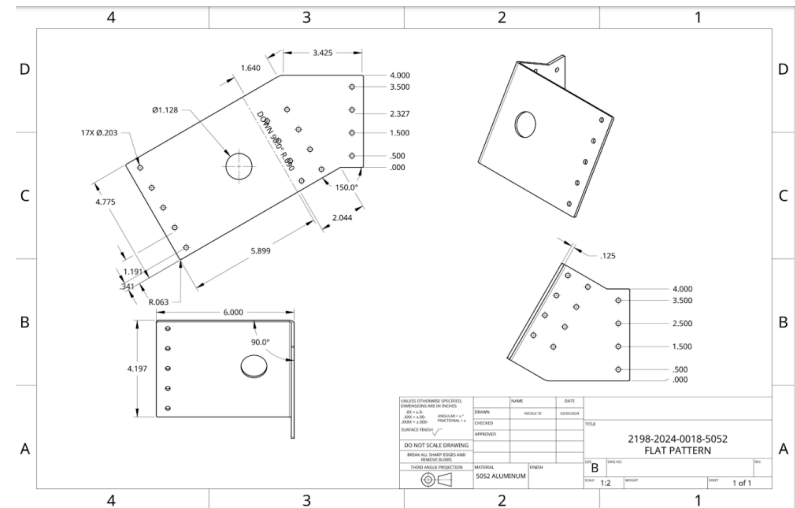
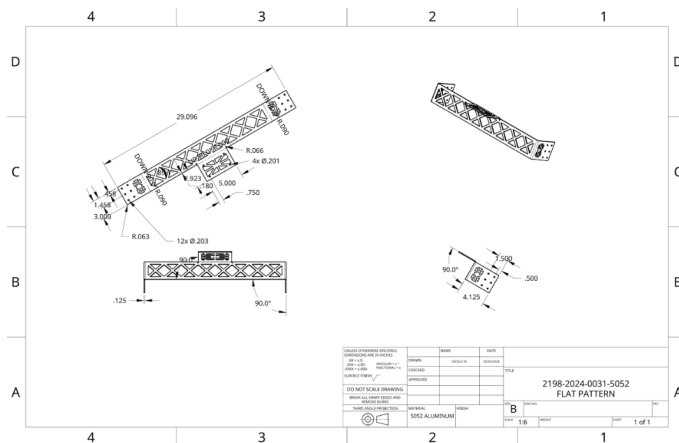
Sep. 2021 - June 2025

- Utilized [TKO Circuit Simulator](#) to design the layout of electrical components 2024-2025 robot
- Designed and implemented a customized electrical system, including wiring, power distribution, and sensor integration, ensuring compliance with competition standards
- Integrated a **line sensor** to detect whether a game piece was secured in the holding mechanism, improving automation
- Wired **encoders** for precise motor control and accurate movement tracking across drive and manipulator systems
- Implemented **color sensors** to differentiate game piece types and enable autonomous decision-making
- Mentored and trained **10+ new members** in electrical wiring, sensor integration, and troubleshooting best practices
- Collaborated with mechanical and programming teams to ensure seamless integration between electrical subsystems



Robot Part Drawings OnShape/Solidworks

- Utilized Solidworks and Onshape to create DXF, STEP, and PDF drawings of sheet metal of First Robotics Robots (2021-2025) for CNC processing
- Designed and fabricated gussets for a FIRST Robotics robot, ensuring precise alignment and secure connections between major mechanisms to enhance structural integrity
- Followed set of tools utilizing standard views, section, and detail dimensions such as model items and smart Dimension, and annotations (including center marks and bill of materials)
- Followed up with metal sponsor with metal parts list organized by part number in folders using Excel



Ecommerce System / Python

- Designed a flowchart to represent the control flow and decision-making process of program
- Integrated key programming concepts like conditional statements, arrays, and math operators to create product sorting , search functionality, and checkout
- Utilized arrays (lists) to store and manage dynamic product data including names and categories
- Implemented bubble sort algorithm to organize product listings by price and rating for user experience
- Designed structured test cases to evaluate different input scenarios and confirm expected program outputs

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Please search below.
Search: fountain pen
Results Of Search

Item Number - 1
Name: wordsworth & black fountain pen set, medium nib
Description: includes 6 ink cartridges and ink refill converter pens [silver gold]
Price: $ 25.0
Rating: 5 / 5

Item Number - 2
Name: jinhao x450 fountain pen (green marble, medium nib 0.7mm)
Description: nib: m (steel nib), cap: push type; converter: screw type
Price: $ 25.5
Rating: 5 / 5

Item Number - 3
Name: arteza - 140-pcs wooden drawing kit
Description: includes 60 crayons, 60 watercolour pans, 10 paintbrushes, 10 hb pencils
Price: $ 40.99
Rating: 5 / 5

Item Number - 4
Name: super squiggles markers - 12 colors self outline markers
Description: double line pen, outline markers pens for art
Price: $ 60.99
Rating: 5 / 5

Item Number - 5
Name: wordsworth & black fountain pen set, medium nib
Description: includes 6 ink cartridges and ink refill converter pens [silver gold]
Price: $ 25.0
Rating: 5 / 5

Would you like to sort items by price, add item to cart, or search for a different item?
1) Enter 1 to sort items by price
2) Enter 2 to add item to cart
3) Enter 3 to Search for a Different Product
Answer: []
    
```

Tutoring Service - Space Academy / JavaScript/HTML/CSS

- Utilized JavaScript to manage cart functionality and total cost calculations
- Created interactive buttons to select courses, add them to a shopping cart, and proceed to checkout
- Designed a clean responsive webpage using HTML and CSS, ensuring a seamless user experience.



CAT'S SPACE ACADEMY

About Us

All Aboard Cat's Space Academy! Whether you're seeking help with math or science, we have all the courses you need. As well, there's a little secret, if you select the 6 pack (All Gr.12 Maths and Sciences), you will receive a 20% discount on your overall purchase! Now click the buttons to select your courses, so I can take off with your money oops I mean to space!:)

My Very Cheap Tutoring Pricing:

Grade 9 Courses	\$30/hr
Grade 10 Courses	\$40/hr
Grade 11 Courses	\$45/hr
Grade 12 Courses	\$50/hr

Math	Science	Cart																
Math 9 Math 10 Functions 11 Data Management 12 Advanced Functions 12 Calculus and Vectors 12	Science 9 Science 10 Chemistry 11 Biology 11 Physics 11 Chemistry 12 Biology 12 Physics 12	<table><thead><tr><th>< Subject ></th><th>< Price ></th></tr></thead><tbody><tr><td>Chemistry 12</td><td>\$50</td></tr><tr><td>Physics 11</td><td>\$45</td></tr><tr><td>Biology 12</td><td>\$50</td></tr><tr><td>Physics 12</td><td>\$50</td></tr><tr><td>Calculus and Vectors 12</td><td>\$50</td></tr><tr><td>Advanced Functions 12</td><td>\$50</td></tr><tr><td>Data Management 12</td><td>\$50</td></tr></tbody></table> 6 Pack Applied: 20% Discount! Subtotal: \$276 Tax (13%): \$ 35.88 Total: \$311.88	< Subject >	< Price >	Chemistry 12	\$50	Physics 11	\$45	Biology 12	\$50	Physics 12	\$50	Calculus and Vectors 12	\$50	Advanced Functions 12	\$50	Data Management 12	\$50
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